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Efficacy of Esbiothrin mosquito coils at community level in northern Tanzania

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Abstract. Field trials with mosquito coils containing 0.15% Esbiothrin at Usa River settlement, northern Tanzania, during 1986 achieved a 52% reduction in biting rates of *Anopheles arabiensis* Patton and a 73% reduction of *Culex quinquefasciatus* Say mosquitoes. No mosquito biting activity occurred during the coil burning period which started at 20.00 hours daily and lasted for an average of 8.6 h. A sociological survey indicated that the coils were favourably accepted and produced no serious side-effects.

Key words. Esbiothrin, mosquito coil, *Anopheles arabiensis*, *Culex quinquefasciatus*, malaria, mosquito biting activity.

Introduction

Mosquito coils containing formulations of insecticides and repellents are often employed for personal protection and to reduce human/vector contact. Assessments of mosquito deterrent effects, mortality rates and feeding inhibition by various types of mosquito coils have been carried out in experimental huts by Smith & Obudho (1967), Feles *et al.* (1968), Hudson & Esozed (1971), Smith *et al.* (1972) and Lubega *et al.* (1977). Results from these studies, as well as from field experiments in Papua New Guinea (Charlwood & Jolly, 1984) and in Kenya (Birley *et al.*, 1984), suggest that mosquito coils are effective in reducing human exposure to endophilic mosquitoes. The present studies were designed to evaluate the efficacy of mosquito coils in reducing human/vector contact when used under normal field conditions at community level in Africa.

Materials and Methods

Studies were conducted at Usa River settlement (altitude 1100 m, 3°45′N 36°85′E), Arusha District, in northern Tanzania. This area is approaching the upper altitudinal limit for endemic malaria and filariasis, on the lower southern flanks between Mount Meru and Mount Kilimanjaro. The settlement consists of three villages, two of which were selected for comparative investigation: Ngarasero (A) and Majengo (B) have, respectively, 144 and 139 houses with 110 and 109 latrines for 1071 and 607 people (census in May 1984). The houses are made of bricks, mud or stick walls, with corrugated iron roofs. The surrounding area has small irrigated paddy and maize farms. Annual rainfall in 1985 totalled 1448 mm with peaks in April–May and November–December. The pre-intervention phase was conducted from December 1984 to January 1985, and the intervention phase of the study was February–April 1986.

Mosquitoes were sampled from four fixed houses in each study village by indoor landing/biting catches on man, conducted twice weekly at 18.00–06.00 hours. During the intervention phase a single mosquito coil containing 0.15% w/w Esbiothrin* was burnt in one room of each of twenty-five houses selected in Majengo village (B) twice weekly. The coils were lit at 20.00 hours and kept smouldering until they burnt out. Surveys of people's opinions were undertaken to measure levels of acceptance, with a view to the future possible use of mosquito coils in malaria control programmes.

* 'Esbiothrin' is the registered trademark of Roussel-Uclaf (France) for the allethrin series pyrethroid with 72% *D-trans* chrysanthemate of *d*-allethrolone as the active ingredient (Wellcome, 1990).

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Results

Mosquito species composition

Culex quinquefasciatus Say and the *Anopheles gambiae* Giles complex were the most abundant species of mos-

quitoes collected indoor-resting and on human bait during the pre-intervention and intervention phase. Several specimens of *An.gambiae sensu lato* processed for cytotaxonomic identification were all found to be *An.arabiensis* Patton. Other species, more abundant outdoors, accounted for less than 2% of the total numbers of mosquitoes collected indoors and included *An.pretoriensis* Giles, *An.funestus s.l.*, *An.coustani* Dönitz, *An.tenebrosus* Dönitz, *Cx duttoni* Theobald, *Cx antennatus* Baker, *Cx annulioris* Theobald, *Cx univittatus* Theobald, *Cx pipiens* L. and *Mansonia uniformis* Theobald. The major breeding site for *Cx quinquefasciatus* was in pit latrines. Densities of this dominant species increased with the rising water level in the pits following heavy rains. *An.arabiensis* was found to breed mainly in rice fields during irrigation and rainy periods.

Mosquito density fluctuations

An.arabiensis was most abundant in January, during the peak of irrigation, and again in March during the early rainy season. Indoors, the mean numbers of bites during these two seasons were 65 and 67 per person per night, respectively. Comparative indoor and outdoor hand collections showed that, during the daytime, *An.arabiensis* was twice as abundant resting outdoors than indoors, with respective mean numbers of 6.3 and 3.6 per man hour (48 man-days of collection). The most productive outdoor resting sites were cattle enclosures. *Cx quinquefasciatus* was present throughout the year, with peak biting activity (46/man-night indoors) after the heavy rains in May.

Mosquito night biting activity

The pre-intervention biting cycles of *An.gambiae s.l.* and *Cx quinquefasciatus* in the untreated village A are shown in Fig. 1. In village B, during the intervention phase when Esbiothrin mosquito coils were utilized, the biting activity declined sharply immediately after the coils started burning (Fig. 1B).

Efficacy of mosquito coils

Esbiothrin mosquito coils ($n = 80$) burned for an average of 8 h 38 min and were quite effective in reducing mosquito bites. In village B, where the coils were burnt, mean biting rates per man-night were 38.6 (12 *An.arabiensis* + 26.6 *Cx quinquefasciatus*) during the pre-intervention phase but only 13 (5.7 *An.arabiensis* + 7.2 *Cx quinquefasciatus*) during the intervention phase, showing reductions of 52% in *An.arabiensis* and 73% in *Cx quinquefasciatus*. During the same period there was no significant change in the densities of mosquitoes in the untreated village A where coils were not burnt (Table 1).

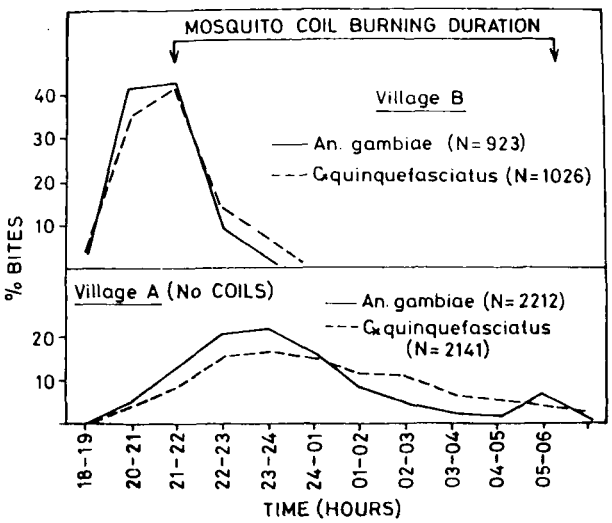


Fig. 1. Nocturnal indoor biting cycle of mosquitoes in the untreated comparison village (A) and treated village (B).

Sociological survey

In the intervention village (B), interviews were conducted between 23 May and 11 June 1986, shortly after completion of the evaluation of the mosquito coils. All the familes indicated their satisfaction with the efficacy of the coils and wanted to continue using them. Among forty-eight heads of families interviewed, 54% had independently used locally manufactured mosquito coils at some time previously. Side-effects, dizziness and running nose, were reported by only 2/48 (4.2%) and these symptoms were mild.

Discussion

These studies indicate that *Cx quinquefasciatus* and *An. arabiensis* were responsible for 99.9% of mosquito bites experienced by residents of villages in the Usa River Settlement area of northern Tanzania during 1985 and 1986. Both species of mosquitoes are important vectors of filariasis in Tanzania and the latter is also a principal malaria vector.

The low proportion of *An.arabiensis* females resting indoors compared to outdoor sites (1:1.75) is characteristic of this sibling species which tends to be more exophilic than *An.gambiae* Giles *sensu strico* or *An.funestus* Giles (Haridi, 1972; White, 1974; Highton *et al.*, 1979). In the apparent absence of these other vector species, *An.arabiensis* is assumed to be the sole malaria vector of importance in the settlement. Malaria is a disease of great concern in the area, accounting for 90.2% ($n = 5242$) of patients attending the local health centre.

In village B where mosquito coils were burnt twice weekly, starting at 20.00 hours and lasting for a mean period of 8.6 h, the overall reduction in mosquito biting rates indoors was 52% *An.arabiensis* and 73% *Cx quinquefasciatus*, comparing the pre-intervention phase (December

Table 1. Mosquito biting densities indoors, with and without the use of mosquito coils, expressed as mean bites per person per night, estimated from human bait collections (18.00–06.00 hours) carried out during the pre-intervention (December 1984 to January 1985) and intervention (February–April 1986) phases of evaluation in treated and untreated villages.

Village	Phase	*An.arabiensis*	*Cx quinquefasciatus*	Other spp.	Total
A, Ngarasero	Pre-intervention	9.2	16.1	0	25.3
(untreated village)	Intervention	11.5	11.2	0.01	22.7
B, Majengo	Pre-intervention	12.0	26.6	0.01	38.6
(treated village)	Intervention	5.7	7.2	0.1	13.0

1984 to June 1985) with the intervention phase (February 1985 to April 1986). No such reduction was observed in village A (Ngarasero) where coils were not burnt systematically (although a few were used spontaneously by the villagers, but not in the houses used for entomological assessment), where the mean density of *An.arabiensis* rose by 25% while that of *Cx quinquefasciatus* fell by 30.4% (Table 1). The long burning period of these Esbiothrin mosquito coils gives effective protection throughout the night period when people are normally asleep. But continued biting by an average of 5.7 *An.arabiensis* and 7.2 *Cx quinquefasciatus* per man-night, mostly prior to lighting the coils at 20.00 hours (Fig. 1B), would be sufficient to maintain disease transmission. It should be beneficial to children, who go to bed earlier, if the coils are lit at sunset prior to the onset of biting activity by endophilic mosquitoes, or if children sleep under mosquito nets. Yet older people are still likely to be exposed during the evening to extra-domiciliary malaria transmission by partially exophagic vectors such as *An.arabiensis*.

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